

nsible for organization unit conventions with collaborative input from the Director/VP/EBA team members for each respective division.

A user can only belong to one role, however they can belong to multiple organization units.

See all of the organization units in the policies and manifests repositories.

Identity Groups

We use Identity Types, Identity Roles and Identity Organization Units across multiple systems for role-based access control (RBAC) by creating an Okta Group, Google Group, and GitLab SaaS Group with the same name. Each of these groups can be attached to one or more Okta Applications, Google Cloud projects or folders, or GitLab Groups or Projects. Additional systems may be added in the future.

Since the user list for each role is centrally managed and users are automatically synced, this removes a lot of manual provisioning inefficiency.

Our automation automatically creates each group and keeps the user list synced every hour with audit logs and automation workflows for any detected attribute and policy manifest changes. See more about manifests, auditlog, and provisioning.

The generated Okta Groups and Google Groups can be assigned to applications, accessed in scheduled generation of JSON/YAML/CSV, accessed via API calls to the Identity Platform, audited by Compliance team members, viewed in generated Google Sheets, and can be trusted as a single source of truth that is updated hourly or daily.

Okta and Google Groups

Each Type group is prefixed with `rbactype{color}`.

Each Organization Unit group is prefixed with `rbacou{ouname}`

Each Role group is prefixed with `rbacrole{color}{rolename}`.

We do not create a role group for brown accounts since each service account has least privilege permissions based on the use case. The grouping of all service accounts uses the `rbactypebrown` group.

GitLab SaaS Groups

The GitLab SaaS group for the type, role, or organization unit can be invited as a Direct Member on any other GitLab group (with child projects) or a GitLab project with the appropriate permission role.

- `@gitlab-rbac/type/{color}`
- `@gitlab-rbac/ou/{ouname}`
- `@gitlab-rbac/role/{rolename}`

This allows us to grant minimum permissions at the top level of the `gitlab-com` and `gitlab-org`

namespaces, and programmatically add elevated Developer, Maintainer, and Owner permissions in downstream groups and projects for specific teams to self manage their own child namespaces without having full read-write access across all projects inherited by the top-level namespace.

In the (distant) future, we will manage GitLab groups and project configuration settings using Terraform.

Identity Policies

Each Identity Role and Identity Organization Unit is defined in a YAML file with a list of conditional rules that defines which user attributes need to match for a user to be added to the manifest.

For granular roles, you can specify one or more job titles. If a job title exists in multiple departments, you can specify job titles in a specific department or that report to a specific manager. For less granular roles, you can use a user's manager since one manager usually owns a specific named function. For larger functional teams, multiple managers can be mapped to a single role.

See all of the policies in the policies repository.

Identity Policy Rulesets

Our rules logic is simple, yet reasonably powerful. You can define one or more attributes to check for, and if any of the attribute conditions match, the user is added to that group.

For programmatic consistency with YAML, all attribute values are converted to lowersnakecase for calculation purposes. This removes all punctuation, replaces spaces with underscores, and solves for slight naming inconsistencies.

plain

Infrastructure => infrastructure

Site Reliability Engineer => sitereliabilityengineer

Director, Infrastructure => directorinfrastructure

Available Attributes

You can use any of the Okta user profile attributes that were imported during the Manifest Users stage.

yaml

```
{accessctkey}: '{oktaApiKey}'
```

emailtohandle:

```
  handle: 'email'
```

```
  manager: 'managerEmail'
```

snakecase:

```
  costcenter: 'costCenter'
```

```
  division: 'division'
```

```
  department: 'department'
```

```
managementlevel: 'workdaymanagementLevel'
organizationname: 'organization'
region: 'workdayregion'
role: 'rbacrole'
title: 'title'
```

Single Attribute Rule

This covers the basic use case of matching the value of a single attribute. If any condition matches, the user will be added to the manifest.

yaml

Any user with these job titles

accountingpayableanalyst:

- title: accountspayableanalyst
- title: senioraccountspayableanalyst

Any user with these job titles

productmanager:

- title: productmanager
- title: seniorproductmanager
- title: principalproductmanager
- title: groupmanagerproduct

Any user that reports to this manager, including the manager too

engproductivity:

- manager: dmurphy
- handle: dmurphy

Multiple Attribute Rule

Since the goal of Identity Groups is to provide granularity, you can define multiple attributes to check (programmatically unlimited, but realistically less than 5). All attribute values must match. To perform an "either/or" rule, you would define an additional rule for each "or".

yaml

Any user in this specific department with this job title

devengleader:

- {department: development, title: distinguishedengineer}
- {department: development, title: seniordistinguishedengineer}
- {department: development, title: engineeringfellow}
- {department: development, title: principalengineer}
- {department: development, title: principalfullstackengineer}

Any user in this department, that is at a specific management level (to accommodate title variations)

devpeopleleader:

- {department: development, managementlevel: vicepresident}

- {department: development, managementlevel: director}

Organization Unit Rules

Each Identity Role policy allows you to use the Okta user profile attribute keys. Organization Unit policies allow you to specify roles as well.

Although it is a best practice to exclusively use roles, we have the flexibility to use attributes or named user handles if needed.

yaml

infrasaaasprodlogviewers:

- department: infrastructure
- handle: dmurphy
- handle: klibby
- role: seclogging
- role: secsirt
- role: engcoredatabase
- role: engcoredbre

yaml

infrasaaasextended:

- {department: infrastructure, managementlevel: leader}
- {department: infrastructure, managementlevel: vicepresident}
- {division: engineering, managementlevel: leader}
- {division: engineering, managementlevel: vicepresident}
- handle: dmurphy
- handle: klibby
- role: infrasaaasleaders
- role: infrasaaasscalabilityleaders
- role: infrasaaasdedicatedgeneral
- role: infrasaaasdedicatedenvautomation
- role: infrasaaasdedicatedswitchboard
- role: infrasaaasdedicateduspubsec
- role: infrasaaasdeployment
- role: infrasaaasfoundations
- role: infrasaaasobservability
- role: infrasaaasops
- role: infrasaaasorchestration
- role: infrasaaaspractices

Manifest Generation

When the Identity Platform CI/CD job runs, it will calculate a fresh list of users that match the rules.

Each rule is evaluated independently, and all user email handles that match that attribute

criteria are included in the returned array.

After all rules have been evaluated, the result arrays from each rule are aggregated and any duplicates are removed if a user matched two or more rules.

plain

Generate a JSON manifest of users based on a policy.yml definition file

Manifest Does Not Exist Yet rbac/roles/{rolename}/manifest.json

```
[member] abarton@example.com Added user to manifest
[member] acummerata@example.com Added user to manifest
[member] akonopelski@example.com Added user to manifest
[member] alegros@example.com Added user to manifest
[member] arunte@example.com Added user to manifest
... truncated
```

[member] Added Total 119 users

File Created rbac/roles/{rolename}/manifest.json

Diff and Drift Management

We have diff detection for the previous and latest manifest for logging and alerting.

plain

Generate a JSON manifest of users based on a policy.yml definition file

```
[member] acummerata@example.com Removed user from manifest
[member] akonopelski@example.com Removed user from manifest
```

[member] Removed Total 2 users

File Updated rbac/roles/{rolename}/manifest.json

Log Schema

<details>

json

```
{
  "message": "User added to manifest",
  "jobbatchid": "{uuid}",
  "timestamp": "YYYY-MM-DDTHH:II:SS.000000Z",
  "event": "accessctl.manifest.create.user.added",
  "policytype": "role",
  "policyname": "{rolename}",
  "manifestemail": "acummerata@example.com",
```

```

"userkey": "acummerata",
"status": "active",
"statusupdatedtimestamp": "YYYY-MM-DDTHH:II:SS.000000Z",
"department": "entr",
"division": "sales",
"title": "insidesalesrepresentative",
"managementlevel": "individualcontributor",
"manager": "vdooley",
"region": "japac",
"rbactype": null,
"rbacrole": "{rolename}"
},
{
  "message": "User removed from manifest",
  "jobbatchid": "{uuid}",
  "timestamp": "YYYY-MM-DDTHH:II:SS.000000Z",
  "event": "accessctl.manifest.create.user.removed",
  "policytype": "role",
  "policyname": "{rolename}",
  "manifestemail": "akonopelski@example.com",
  "userkey": null,
  "status": "deprovisioned",
  "statusupdatedatiso8601": "YYYY-MM-DDTHH:II:SS.000000Z"
}

```

</details>

Group User Sync

Many vendors rules engines work well for adding a user to a group, but there are limitations with removing a user from the group. This problem is amplified with offboarded users since deactivated users aren't removed from the groups automatically (their authentication is blocked, however their authorization is still in-tact).

There is also limited to no functionality available with vendors to detecting changes and diff logging or triggering automation workflows.

With accessctl, we have advanced change/diff audit logging and workflow automation, and keep groups in sync for adding and removing users.

For any Google Groups, Okta Groups, or tech stack APIs that are managed by the Identity Platform, our CI/CD jobs query the API to get all current members for that group, compare it to our manifest, and perform add user or remove user API calls if needed.

Adding Users to Group

php

```

// https://developer.okta.com/docs/reference/api/groups/add-user-to-group
$oktaresponse = OktaApiClient::post(

```

```

        connection: config('connections.okta.prod'),
        uri: 'groups/ ' . $oktagroupid . '/members/' . $oktauserid,
    );

// https://developers.google.com/admin-sdk/directory/reference/rest/v1/members/insert
$googleresponse = GoogleApiClient::post(
    connection: config('connections.google.workspace'),
    url: 'https://admin.googleapis.com/admin/directory/v1/groups/' . $googlegroupid .
'/members',
    formdata: [
        'email' => $email,
        'role' => 'MEMBER',
        'type' => 'USER',
    ]
);

// https://docs.gitlab.com/ee/api/members.htmladd-a-member-to-a-group-or-project
// https://docs.gitlab.com/ee/api/members.htmlroles
$gitlabresponse = GitlabApiClient::post(
    connection: config('connections.gitlab.saas'),
    uri: 'groups/' . $gitlabgroupid . '/members',
    requestdata: [
        'userid' => $gitlabuserid,
        'accesslevel' => $gitlabaccesslevel,
        'expiresat' => null
    ]
);

```

Removing Users from Group

```

php
// https://developer.okta.com/docs/reference/api/groups/remove-user-from-group
$oktaresponse = OktaApiClient::delete(
    connection: config('connections.okta.prod'),
    uri: 'groups/ ' . $oktagroupid . '/members/' . $oktauserid,
);

// https://developers.google.com/admin-sdk/directory/reference/rest/v1/members/delete
$googleresponse = GoogleApiClient::delete(
    connection: config('connections.google.workspace'),
    url: 'https://admin.googleapis.com/admin/directory/v1/groups/' . $googlegroupid .
'/members/' . $email
);

// https://docs.gitlab.com/ee/api/members.htmlremove-a-member-from-a-group-or-project
$gitlabresponse = GitlabApiClient::delete(
    connection: config('connections.gitlab.saas'),
    uri: 'groups/' . $gitlabgroupid . '/members/' . $gitlabuserid
);

```

CI/CD Pipeline Jobs

We consider Workday to be the source of truth for new users and changes to user profile attribute values, and we fetch those values from the Okta API Users endpoint. When a user's Okta status changes to deprovisioned, we consider them to be offboarded and remove them from all manifests.

The Identity Platform runs scheduled CI/CD pipeline jobs that use accessctl CLI jobs to recalculate the list of users (a.k.a. manifest) for every Identity Role and Identity Organization Unit based on the latest data.

When a user attribute value changes in Workday and pushes to Okta, the Identity Platform will detect the diff during the manifests pipeline stage and create audit log entries and dispatch automation workflows in the auditlog pipeline stage. After a manifest has been created, the provisioning pipeline syncs all of the groups and group users across the rbac prefixed Okta groups, Google Groups, @gitlab-rbac/ GitLab groups that the Identity Platform manages.

When we reach maturity, the scheduled jobs will run every hour and all attribute values and groups will be automatically calculated and programatically maintained without any human user intervention or day-to-day provisioning tasks. While we are in PoC/alpha/beta, the jobs run daily and created in a merge request that is reviewed by the Identity Ops team for accuracy to catch any bugs or unintended changes.

mermaid
graph LR

subgraph accessctl GitLab CI/CD Pipeline Jobs

direction LR

subgraph Manifest Stage

direction LR

CIMANIFESTUSERJOB["Stage 1.1
Users Job
CLI manifest:users"]

CIMANIFESTROLEJOB["Stage 1.2
Roles Job
CLI manifest:roles"]

CIMANIFESTGROUPJOB["Stage 1.3
Org Units Job
CLI manifest:ou"]

end

subgraph Auditlog Stage

direction LR

CICHANGELOGUSERJOB["Stage 2.1
Users Job
CLI auditlog:users"]

CICHANGELOGATTRIBUTEJOB["Stage 2.2
Attributes Job
CLI auditlog:attributes"]

CICHANGELOGROLEJOB["Stage 2.3
Roles Job
CLI auditlog:roles"]

CICHANGELOGGROUPJOB["Stage 2.4
Org Units Job
CLI auditlog:ou"]

end

subgraph Provisioning Stage

direction LR

subgraph Okta API

CIPROVISIONINGOKTAUSERJOB["Stage 3.1
Okta Users Job
provision:okta-users"]

CIPROVISIONINGOKTAGROUPJOB["Stage 3.2
Okta Groups Job
provision:okta-groups"]

end

subgraph Google Workspace Directory API

CIPROVISIONINGGOOGLEGROUPJOB["Stage 3.3
Google Groups Job
provision:google-groups"]

```

end
subgraph GitLab.com SaaS API
CIPROVISIONINGGITLABSAASGROUPJOB["Stage 3.4<br />GitLab SaaS Groups Job<br />provision:gitla
saas-groups"]
end
subgraph GitLab Self-Managed Instance API
CIPROVISIONINGGITLABSELFGROUPJOB["Stage 3.5<br />GitLab Self-Managed Groups Job<br
/>provision:gitlab-self-groups"]
end
end
end

```

```

CIMANIFESTUSERJOB --> CIMANIFESTROLEJOB --> CIMANIFESTGROUPJOB
CICHANGELOGUSERJOB --> CICHANGELOGATTRIBUTEJOB --> CICHANGELOGROLEJOB --> CICHANGELOGGROUPJOB
CIPROVISIONINGOKTAUSERJOB --> CIPROVISIONINGOKTAGROUPJOB --> CIPROVISIONINGGOOGLEGROUPJOB
CIPROVISIONINGGITLABSAASGROUPJOB --> CIPROVISIONINGGITLABSELFGROUPJOB

```

```

click CIMANIFESTUSERJOB "/handbook/security/identity/platform/manifests" "View Details"
click CIMANIFESTROLEJOB "/handbook/security/identity/platform/manifests" "View Details"
click CIMANIFESTGROUPJOB "/handbook/security/identity/platform/manifests" "View Details"
click CICHANGELOGUSERJOB "/handbook/security/identity/platform/auditlog" "View Details"
click CICHANGELOGATTRIBUTEJOB "/handbook/security/identity/platform/auditlog" "View Details"
click CICHANGELOGROLEJOB "/handbook/security/identity/platform/auditlog" "View Details"
click CICHANGELOGGROUPJOB "/handbook/security/identity/platform/auditlog" "View Details"
click CIPROVISIONINGOKTAUSERJOB "/handbook/security/identity/platform/provisioning/okta" "View
Details"
click CIPROVISIONINGOKTAGROUPJOB "/handbook/security/identity/platform/provisioning/okta" "View
Details"
click CIPROVISIONINGGOOGLEGROUPJOB "/handbook/security/identity/platform/provisioning/google"
"View Details"
click CIPROVISIONINGGITLABSAASGROUPJOB
"/handbook/security/identity/platform/provisioning/gitlab" "View Details"
click CIPROVISIONINGGITLABSELFGROUPJOB
"/handbook/security/identity/platform/provisioning/gitlab" "View Details"

```

```

classDef slate fill:cbd5e1,stroke:475569,stroke-width:1px;
classDef red fill:fca5a5,stroke:dc2626,stroke-width:1px;
classDef orange fill:fd8b74,stroke:ea580c,stroke-width:1px;
classDef yellow fill:fcd34d,stroke:ca8a04,stroke-width:1px;
classDef emerald fill:6ee7b7,stroke:059669,stroke-width:1px;
classDef cyan fill:67e8f9,stroke:0891b2,stroke-width:1px;
classDef sky fill:7dd3fc,stroke:0284c7,stroke-width:1px;
classDef violet fill:c4b5fd,stroke:7c3aed,stroke-width:1px;
classDef fuchsia fill:f0abfc,stroke:c026d3,stroke-width:1px;
end

```

Configuration and Data Repositories

The tf Terraform GitOps repository contains the Terraform that manages the configuration and resources in each platform/vendor (ex. Okta Settings and Okta Applications), and specifies

which groups are assigned to each application/resource/permission in each respective platform (ex. Okta application groups, GCP project roles, etc). We believe that configuration and state management auditlogs are much easier to audit without all of the user membership changes, so we use Access Control to keep those changes separate and provide improved automation user experience for those use cases. In other words, it's not realistic to have every user perform self-service Terraform changes for access requests.

Access Control uses the Identity Data policies and manifests repositories that are focused on who belongs to a group.

The policies repository is for managing YAML policy files that are managed by humans using the CLI for generating properly formatted changes and has strict merge request approval rules for change management before changes are merged.

The manifests repository has the data that has been generated by the accessctl automation, with detected changes separately added in the auditlog repository for audit preservation.

See the mirrored PoC data repositories for a sneak peak of the future of Identity v3.

mermaid
graph LR

subgraph accessctl GitLab Repositories

direction LR

subgraph accessctl-policies Repo

direction TB

REPOINVPOLICIESROLES[("policies/role/{kingdom}.yaml")]

REPOINVPOLICIESOU[("policies/ou/{kingdom}.yaml")]

end

subgraph accessctl-manifests Repo

direction TB

REPOINVMANIFESTSUSERS[("manifests/users/
users.yaml/json/csv")]

REPOINVMANIFESTSATTRIBUTES[("manifests/attributes/
{attribute}.yaml/json/csv")]

REPOINVMANIFESTSROLE[("manifests/roles/
{role}.yaml/json/csv")]

REPOINVMANIFESTSOU[("manifests/ou/
{ou}.yaml/json/csv")]

end

subgraph accessctl-auditlog Repo

direction TB

REPOINVCHANGELOGUSERSONBOARDING[("auditlog/users/
onboarding.yaml/json/csv")]

REPOINVCHANGELOGUSERSOFFBOARDING[("auditlog/users/
offboarding.yaml/json/csv")]

REPOINVCHANGELOGUSERSCHANGES[("auditlog/users/
attributes.yaml/json/csv")]

REPOINVCHANGELOGATTRIBUTE[("auditlog/attribute/
{attribute}.yaml/json/csv")]

REPOINVCHANGELOGROLE[("auditlog/role/
{role}.yaml/json/csv")]

REPOINVCHANGELOGOU[("auditlog/ou/
{ou}.yaml/json/csv")]

end

end

classDef slate fill:cbd5e1,stroke:475569,stroke-width:1px;

classDef red fill:fca5a5,stroke:dc2626,stroke-width:1px;

classDef orange fill:fdba74,stroke:ea580c,stroke-width:1px;

```

classDef yellow fill:fcd34d,stroke:ca8a04,stroke-width:1px;
classDef emerald fill:6ee7b7,stroke:059669,stroke-width:1px;
classDef cyan fill:67e8f9,stroke:0891b2,stroke-width:1px;
classDef sky fill:7dd3fc,stroke:0284c7,stroke-width:1px;
classDef violet fill:c4b5fd,stroke:7c3aed,stroke-width:1px;
classDef fuchsia fill:f0abfc,stroke:c026d3,stroke-width:1px;

```

Data Security

- Although these could be part of the same repository, we provide separation so that the pipeline automation has read-only access to the policies repository to avoid any potential compromise.
- The SSOT repository is hosted on the Black OPs self-managed GitLab instance to protect the data integrity, and repository data is mirrored to GitLab.com SaaS for team member read-only visibility.
- The runtime pipelines, CI/CD variables, and secrets management are on the Black Ops GitLab instance that no team members have elevated permissions on other than break glass access by the Identity Engineering Team. We use the CLI for generating properly formatted changes via API calls to avoid users making freeform changes or clicking buttons.

Data Flow

```

mermaid
graph TB

```

```

subgraph "Workday"
    WORKDAYWORKER[Worker]
    WORKDAYREGION([Region])
    WORKDAYDEPARTMENT([Department])
    WORKDAYDIVISION([Division])
    WORKDAYJOBTITLE([Job Title])
    WORKDAYDEPARTMENT -.- WORKDAYWORKER
    WORKDAYDIVISION -.- WORKDAYWORKER
    WORKDAYREGION -.- WORKDAYWORKER
    WORKDAYJOBTITLE -.- WORKDAYWORKER
    WORKDAYWORKER -. "Manager Relationship" .-> WORKDAYWORKER
end

```

```

subgraph Identity SaaS Vendor Services
    direction LR
    subgraph Okta GitLab Tenant
        OKTAPROFILEMAPPING{{Workday to Okta<br>Profile Mapping}}
        IDENTITYVENDOROKTAAPIUSERS(["Okta Users"])
        IDENTITYVENDOROKTAAPIENDPOINT(["Okta API<br />https://gitlab.okta.com/api/v1"])
        WORKDAYWORKER -- "SCIM Provisioning<br />Every Hour" --> OKTAPROFILEMAPPING
        OKTAPROFILEMAPPING --> IDENTITYVENDOROKTAAPIUSERS
        IDENTITYVENDOROKTAAPIUSERS --> IDENTITYVENDOROKTAAPIENDPOINT
    end
end

```

```

subgraph Automation Plane
subgraph Access Control CI/CD Pipeline
direction LR
CISTAGE1MANIFEST["Manifest Stage 1"]
CISTAGE2CHANGELOG["Auditlog Stage 2"]
CISTAGE3PROVISIONING["Provisioning Stage 3"]
subgraph REST API Client
direction LR
CIRESTINTEGRATION["Access Control<br />Integration API"]
CIRESTOKTA["Okta API<br />Groups"]
CIRESTGOOGLE["Google API<br />Group Users"]
CIRESTGITLAB["GitLab API"]
end
end

```

```

subgraph Okta API
OKTAGROUPS["Okta rbac Groups"]:::emerald
OKTAGROUPSUSERS["Group Users"]:::emerald
OKTAUSERS["Okta Users"]
OKTAGROUPS -. OKTAGROUPSUSERS
OKTAGROUPSUSERS -. OKTAUSERS
end

```

```

subgraph Google Workspace Directory API
GOOGLEGROUPS["Google rbac Groups"]:::emerald
GOOGLEGROUPSUSERS["Group Members"]:::emerald
GOOGLEUSERS["Google Users"]
GOOGLEGROUPS -. GOOGLEGROUPSUSERS
GOOGLEGROUPSUSERS -. GOOGLEUSERS
end

```

```

subgraph GitLab API for SaaS and ops.gitlab.net
GITLABGROUPS["@gitlab-rbac/ Groups"]:::orange
GITLABGROUPSUSERS["Group Members"]:::orange
GITLABUSERS["@gitlab.com Users"]
GITLABGROUPS -. GITLABGROUPSUSERS
GITLABGROUPSUSERS -. GITLABUSERS
end
end

```

```

subgraph Control Plane
subgraph Slack
CTLSLACK["/accessctl<br />Slack Bot<br />End Users and Approvers"]:::red
end
subgraph Mac and Linux
CTLCLI["accessctl CLI<br />Power Users and Admins"]:::red
end
subgraph Web Application
CTLUI["ctl.gitlab.systems<br/>accessctl Identity UI<br />Team Member Self Service"]:::red
end

```

end

subgraph Data Plane

direction TB

CTLAPI["ctl.gitlab.black
accessctl Identity API
(behind Teleport bastion)"]:::re

CTLAPI -- REPOPOLICIES

CTLAPI -- REPOMANIFESTS

CTLAPI -- REPOCHANGELOG

subgraph Access Control Repositories

REPOPOLICIES["accessctl-policies Repo
policies/.yaml"]

REPOMANIFESTS["accessctl-manifests Repo
manifests/.yaml/json/csv"]

REPOCHANGELOG["accessctl-auditlog Repo
auditlog/.yaml/json/csv"]

end

CTLS3["Access Control S3 Logs
and Repo Backup"]

SIEMDEVOS3["Devo ingests S3 logs"]:::violet

end

subgraph Integration Control Plane

subgraph Access Control Integration Scripts

INTEGRATIONAPI["Integration API/Webhooks"]

INTEGRATIONBLUEPRINT["Blueprints (Future)
Script Templates to replace
no code use cases for
common REST API calls
(Email, Slack Post, Create Issue,
Add Row to Sheet, etc)"]

INTEGRATIONTRANSACTION["Database Audit Transaction
Database rows for each event
with metadata for blueprint variables
and dispatching jobs on Redis queue."]

INTEGRATIONJOB["Queued Jobs"]

INTEGRATIONTRANSACTION -- INTEGRATIONBLUEPRINT

INTEGRATIONBLUEPRINT -- INTEGRATIONJOB

INTEGRATIONTRANSACTION --> INTEGRATIONJOB

end

INTEGRATIONLEGACYARISSUE["Auto-Generated AR Issue
for legacy audit logging
transition in FY25"]:::violet

INTEGRATIONGOOGLESHEET["Google Sheets
Audit Inventory Reports"]:::violet

INTEGRATIONSLACKNOTIFICATION["Slack Notifications
to Channel(s) or User(s)"]:::violet

end

IDENTITYVENDOROKTAAPIENDPOINT --> CISTAGE1MANIFEST

CTLCLI --> CTLAPI

CTLSLACK --> CTLAPI

CTLUI --> CTLAPI

CTLS3 --> SIEMDEVOS3

REPOPOLICIES --> CISTAGE1MANIFEST

CISTAGE2CHANGELOG --> REPOCHANGELOG

CISTAGE2CHANGELOG --> CTLS3

CISTAGE3PROVISIONING --> CTLS3

REPOMANIFESTS --> CISTAGE3PROVISIONING

CISTAGE1MANIFEST --> CISTAGE2CHANGELOG

CISTAGE1MANIFEST --> REPOMANIFESTS

REPOMANIFESTS --> CISTAGE2CHANGELOG

CISTAGE2CHANGELOG --> CISTAGE3PROVISIONING

CISTAGE3PROVISIONING -----> CIRESTOKTA
CISTAGE3PROVISIONING --> CIRESTGOOGLE
CISTAGE3PROVISIONING --> CIRESTGITLAB
CISTAGE2CHANGELOG ----> CIRESTINTEGRATION
CISTAGE3PROVISIONING ----> CIRESTINTEGRATION
CIRESTINTEGRATION -----> INTEGRATIONAPI
INTEGRATIONAPI --> INTEGRATIONJOB
INTEGRATIONJOB --> INTEGRATIONLEGACYARISSUE
INTEGRATIONJOB --> INTEGRATIONGOOGLESHEET
INTEGRATIONJOB --> INTEGRATIONSLACKNOTIFICATION
CIRESTOKTA --> OKTAGROUPSUSERS
CIRESTOKTA --> OKTAGROUPS
CIRESTGOOGLE --> GOOGLEGROUPSUSERS
CIRESTGOOGLE --> GOOGLEGROUPS
CIRESTGITLAB --> GITLABGROUPS
CIRESTGITLAB --> GITLABGROUPSUSERS

```
classDef slate fill:cbd5e1,stroke:475569,stroke-width:1px;
classDef red fill:fca5a5,stroke:dc2626,stroke-width:1px;
classDef orange fill:fdba74,stroke:ea580c,stroke-width:1px;
classDef yellow fill:fcd34d,stroke:ca8a04,stroke-width:1px;
classDef emerald fill:6ee7b7,stroke:059669,stroke-width:1px;
classDef cyan fill:67e8f9,stroke:0891b2,stroke-width:1px;
classDef sky fill:7dd3fc,stroke:0284c7,stroke-width:1px;
classDef violet fill:c4b5fd,stroke:7c3aed,stroke-width:1px;
classDef fuchsia fill:f0abfc,stroke:c026d3,stroke-width:1px;
```

Access Control (accessctl)

Access Control is an open source monolith code base created by Jeff Martin that is built with the Laravel framework with multiple functional components included that are deployed in a distributed fashion to dogfood GitLab CI/CD pipelines and GitLab Runner for background jobs, GitLab repository for policy and manifest storage, CODEOWNERS and merge request approvals for change management.

Although Laravel has batteries included for all of the full stack web application and SQL database capabilities that rivals Django or Ruby on Rails, we are using it as more of a scripting engine that is easier for team members to contribute to and leverages GitLab for backend processing wherever possible. We still use Laravel native capabilities to expose a REST API, Web UI, and CLI for improved frontend user experience.

We use Laravel Actions for our business logic service classes. By using Laravel Actions, we simplify our architecture since each single responsibility principle (SRP) action can be invoked as a class, a console CLI command, or a background job. This also allows us to get data from the GitLab API for repository data instead of needing to use models and migrations for a large SQL database.

See the [gitlab-identity/accessctl](#) repository and README to learn more.

Why Laravel

It's a simple case of using the tools that we know and can deliver business value efficiently. Jeff Martin is the primary developer has been using it for 10+ years and has tried using Ruby on Rails, Golang, Python, and still finds that the Laravel ecosystem is richer.

You may have already seen what's possible with GitLab Sandbox Cloud, Demo Systems, and Training Lab Manager. It just works and only requires 1 engineer part time to take us to production. We've taught more than a dozen team members how Laravel works without any trouble. This isn't part of the GitLab product itself so that's a non factor. Please reach out to Jeff for a code pair if you're interested.

CI/CD Scripts

We use a Docker image with the accessctl application with GitLab Runner to run Artisan Console commands that from Laravel actions. Each CI/CD job is usually related to ETL operations for (E)xtracting IAM/RBAC data from vendor APIs, (T)ransforming the API responses into standardized YAML and JSON array schemas, and (L)oding the data into the GitLab repositories (accessctl-manifests, accessctl-auditlog and accessctl-policies for system automation) and Google Sheets (for human use).

The CI/CD pipeline has several stages:

- Manifests Stage
- Auditlog Stage
- Provisioning Stage

API

We have the accessctl application deployed at <https://ctl.gitlab.black> that provides a REST API for data stored in the GitLab repository that is proxied and formatted as if we used a traditional SQL database. This improves the consumability of our data, and provides an endpoint for CLI application.

This is deployed separately from the Web UI since we use network ACLs to only allow access for users connected to the VPN or service accounts that have pre-configured allowlist IP addresses.

Web UI

We have the accessctl application deployed at <https://ctl.gitlab.systems> that provides Web UI with Okta authentication (with 2FA and Okta Verify device trust) for end user team members and managers to perform day-to-day tasks without administrative access.

For security and UI design internal tool efficiency reasons, only the CLI is used for admin actions.

Slack Bot

We will be building a Slack bot that will be used for end user form wizard workflows and just-in-time/manager approvals to supplement the Web UI.

CLI

The accessctl CLI is used by power users and admins for generating properly formatted MRs with policy changes with the accessctl API.

We only accept system generated changes. No human generated (manual edit) MRs are allowed.

Access Check (accesschk)

Access Check is a series of scripts for performing audit evidence collection. It is part of the accessctl codebase but is run using an independent CI/CD pipeline.

You can think of this as an audit automation platform that also provides a single source of truth (SSOT) data source for the latest aggregated information available from each vendor's API and saving it as a CSV/JSON/YML file and Google Sheet that can be referenced by auditors, compliance teams, and other team members as needed.

See the accesschk handbook page to learn more.

Additional Reading

See the Identity Platform Architecture and User Guide links at the top of this page.

SECTION: security/identity/platform/accesschk/_index.md

{{% alert title="Not Live Yet" color="warning" %}}

You are viewing a preview of documentation for the future state of GitLab Identity v3 (mid 2024). See the Access Management Policy for the GitLab Identity v2 current state with baseline entitlements and access requests. See the roadmap in the epics gantt chart.

{{% /alert %}}

{{% alert title="Not the documentation you are looking for?" color="info" %}}

You are viewing the accesschk engineering deep-dive architecture for audit and compliance evidence collection. We have accessctl engineering architecture documentation for policy management and automated provisioning. We also have a getting started guide for auditors, change management, and tech stack application system owners.

{{% /alert %}}

{{% alert title="Work in Progress" color="danger" %}}

This page is a work-in-progress. Please check back later for up-to-date details.

{{% /alert %}}

CI/CD Pipeline Overview

mermaid
graph LR

```
subgraph accesschk GitLab Repositories
direction LR
subgraph accesschk-evidence Repo
direction LR
end
end
```

```
subgraph accesschk GitLab CI/CD Pipeline Jobs
```

```
direction LR
```

```
subgraph Okta API
```

```
CIAUDITOKTAUSERJOB["Okta Users Job<br />chk:okta-users"]
```

```
CIAUDITOKTAAPPJOB["Okta Apps Job<br />chk:okta-apps"]
```

```
CIAUDITOKTAGROUPJOB["Okta Groups Job<br />chk:okta-groups"]
```

```
CIAUDITOKTAPOLICYJOB["Okta Policies Job<br />chk:okta-policies"]
```

```
CIAUDITOKTAADMINROLESJOB["Okta Admin Roles Job<br />chk:okta-admin-roles"]
```

```
CIAUDITOKTASETJOB["Okta Settings Job<br />chk:okta-settings"]
```

```
end
```

```
subgraph Google Workspace Directory API
```

```
CIAUDITGOOGLEUSERJOB["Google Users Job<br />chk:google-users"]
```

```
CIAUDITGOOGLEADMINROLESJOB["Google Admin Roles Job<br />chk:google-admin-roles"]
```

```
CIAUDITGOOGLEGROUPJOB["Google Groups Job<br />chk:google-groups"]
```

```
CIAUDITGOOGLECHROMEJOB["Google Chrome Policies Job<br />chk:google-chrome"]
```

```
CIAUDITGOOGLEORGUNITJOB["Google Org Units Job<br />chk:google-org-units"]
```

```
end
```

```
subgraph Google Cloud Resource Manager and IAM API
```

```
CIAUDITGCPORGSJOB["Google Cloud Organizations Job<br />chk:gcp-organizations"]
```

```
CIAUDITGCPFOLDERSJOB["Google Cloud Folders Job<br />chk:gcp-folders"]
```

```
CIAUDITGCPPROJECTSJOB["Google Cloud Projects Job<br />chk:gcp-projects"]
```

```
CIAUDITGCPSERVICEACCOUNTSJOB["Google Cloud Service Accounts Job<br />chk:gcp-service-accounts"]
```

```
CIAUDITGCPBILLINGACCOUNTSJOB["Google Cloud Billing Accounts Job<br />chk:gcp-billing-accounts"]
```

```
end
```

```
subgraph GitLab.com SaaS API
```

```
CIAUDITGITLABSAASGROUPJOB["GitLab SaaS Groups Job<br />chk:gitlab-saas-groups"]
```

```
CIAUDITGITLABSAASPROJECTSJOB["GitLab SaaS Projects Job<br />chk:gitlab-saas-projects"]
```

```
CIAUDITGITLABSAASADMINJOB["GitLab SaaS Admin Roles Job<br />chk:gitlab-saas-admins"]
```

```
end
```

```
subgraph GitLab Self-Managed Instance API
```

```
CIAUDITGITLABSELFJOB["GitLab Self-Managed Groups Job<br />chk:gitlab-self-groups"]
```

```
CIAUDITGITLABSELFPROJECTSJOB["GitLab Self-Managed Projects Job<br />chk:gitlab-self-projects"]
```

```
CIAUDITGITLABSELFADMINJOB["GitLab Self-Managed Admin Roles Job<br />chk:gitlab-self-admins"]
```

```
end
```

```
end
```

SECTION: security/identity/platform/auditlog/_index.md

{{% alert title="Not Live Yet" color="warning" %}}

You are viewing a preview of documentation for the future state of GitLab Identity v3 (mid 2024). See the [Access Management Policy](/handbook/security/access-management-policy) for the GitLab Identity v2 current state with baseline entitlements and access requests. See the roadmap in the [epics gantt chart](https://gitlab.com/groups/gitlab-com/gi-security/identity/eng/-/roadmap?state=all&sort=startdateasc&layout=QUARTERS&timeframerangetype=THREEYEARS&group=gitlab-com/gi-security/identity/eng&progress=WEIGHT&showprogress=true&showmilestones=false&milestonetype=ALL&showlabels=true).

{{% /alert %}}

Pipeline Overview

mermaid
graph LR

subgraph accessctl GitLab Repositories

direction LR

subgraph accessctl-policies Repo

direction TB

REPOINVPOLICIESROLES[("policies/role/{kingdom}.yaml")]

REPOINVPOLICIESOU[("policies/ou/{kingdom}.yaml")]

end

subgraph accessctl-manifests Repo

direction TB

REPOINVMANIFESTSUSERS[("manifests/users/
users.yaml/json/csv")]:::sky

REPOINVMANIFESTSATTRIBUTES[("manifests/attributes/
{attribute}.yaml/json/csv")]:::sky

REPOINVMANIFESTSROLE[("manifests/roles/
{role}.yaml/json/csv")]:::sky

REPOINVMANIFESTSOU[("manifests/ou/
{ou}.yaml/json/csv")]:::sky

end

subgraph accessctl-auditlog Repo

direction TB

REPOINVCHANGELOGUSERSONBOARDING[("auditlog/users/
onboarding.yaml/json/csv")]:::emerald

REPOINVCHANGELOGUSERSOFFBOARDING[("auditlog/users/
offboarding.yaml/json/csv")]:::emerald

REPOINVCHANGELOGUSERSCHANGES[("auditlog/users/
attributes.yaml/json/csv")]:::emerald

REPOINVCHANGELOGATTRIBUTE[("auditlog/attribute/
{attribute}.yaml/json/csv")]:::emerald

REPOINVCHANGELOGROLE[("auditlog/role/
{role}.yaml/json/csv")]:::emerald

REPOINVCHANGELOGOU[("auditlog/ou/
{ou}.yaml/json/csv")]:::emerald

end

end

subgraph accessctl GitLab CI/CD Pipeline Jobs

direction LR

subgraph Manifest Stage

direction LR

CIMANIFESTUSERJOB[["Stage 1.1
Users Job
CLI manifest:users"]]

```

CIMANIFESTROLEJOB["Stage 1.2<br />Roles Job<br />CLI manifest:roles"]
CIMANIFESTGROUPJOB["Stage 1.3<br />Org Units Job<br/>CLI manifest:ou"]
end
subgraph Auditlog Stage
direction LR
CICHANGELOGUSERJOB["Stage 2.1<br />Users Job<br/>CLI auditlog:users"]:::orange
CICHANGELOGATTRIBUTEJOB["Stage 2.2<br />Attributes Job<br/>CLI auditlog:attributes"]:::orange
CICHANGELOGROLEJOB["Stage 2.3<br />Roles Job<br/>CLI auditlog:roles"]:::orange
CICHANGELOGGROUPJOB["Stage 2.4<br />Org Units Job<br/>CLI auditlog:ou"]:::orange
end
subgraph Provisioning Stage
direction LR
subgraph Okta API
CIPROVISIONINGOKTAUSERJOB["Stage 3.1<br />Okta Users Job<br />provision:okta-users"]
CIPROVISIONINGOKTAGROUPJOB["Stage 3.2<br />Okta Groups Job<br />provision:okta-groups"]
end
subgraph Google Workspace Directory API
CIPROVISIONINGGOOGLEGROUPJOB["Stage 3.3<br />Google Groups Job<br />provision:google-groups"]
end
subgraph GitLab.com SaaS API
CIPROVISIONINGGITLABSAASGROUPJOB["Stage 3.4<br />GitLab SaaS Groups Job<br />provision:gitlab-saas-groups"]
end
subgraph GitLab Self-Managed Instance API
CIPROVISIONINGGITLABSELFGROUPJOB["Stage 3.5<br />GitLab Self-Managed Groups Job<br />provision:gitlab-self-groups"]
end
end

```

```

CIMANIFESTUSERJOB --> CIMANIFESTROLEJOB --> CIMANIFESTGROUPJOB
CICHANGELOGUSERJOB --> CICHANGELOGATTRIBUTEJOB --> CICHANGELOGROLEJOB --> CICHANGELOGGROUPJOB
CIPROVISIONINGOKTAUSERJOB --> CIPROVISIONINGOKTAGROUPJOB --> CIPROVISIONINGGOOGLEGROUPJOB
CIPROVISIONINGGITLABSAASGROUPJOB --> CIPROVISIONINGGITLABSELFGROUPJOB

```

```

click CIMANIFESTUSERJOB "/handbook/security/identity/platform/manifests" "View Details"
click CIMANIFESTROLEJOB "/handbook/security/identity/platform/manifests" "View Details"
click CIMANIFESTGROUPJOB "/handbook/security/identity/platform/manifests" "View Details"
click CICHANGELOGUSERJOB "/handbook/security/identity/platform/auditlog" "View Details"
click CICHANGELOGATTRIBUTEJOB "/handbook/security/identity/platform/auditlog" "View Details"
click CICHANGELOGROLEJOB "/handbook/security/identity/platform/auditlog" "View Details"
click CICHANGELOGGROUPJOB "/handbook/security/identity/platform/auditlog" "View Details"
click CIPROVISIONINGOKTAUSERJOB "/handbook/security/identity/platform/provisioning/okta" "View Details"
click CIPROVISIONINGOKTAGROUPJOB "/handbook/security/identity/platform/provisioning/okta" "View Details"
click CIPROVISIONINGGOOGLEGROUPJOB "/handbook/security/identity/platform/provisioning/google" "View Details"
click CIPROVISIONINGGITLABSAASGROUPJOB
"/handbook/security/identity/platform/provisioning/gitlab" "View Details"
click CIPROVISIONINGGITLABSELFGROUPJOB

```

["/handbook/security/identity/platform/provisioning/gitlab"](/handbook/security/identity/platform/provisioning/gitlab) "View Details"

```
classDef slate fill:cbd5e1,stroke:475569,stroke-width:1px;
classDef red fill:fca5a5,stroke:dc2626,stroke-width:1px;
classDef orange fill:fdba74,stroke:ea580c,stroke-width:1px;
classDef yellow fill:fcd34d,stroke:ca8a04,stroke-width:1px;
classDef emerald fill:6ee7b7,stroke:059669,stroke-width:1px;
classDef cyan fill:67e8f9,stroke:0891b2,stroke-width:1px;
classDef sky fill:7dd3fc,stroke:0284c7,stroke-width:1px;
classDef violet fill:c4b5fd,stroke:7c3aed,stroke-width:1px;
classDef fuchsia fill:f0abfc,stroke:c026d3,stroke-width:1px;
end
```

CI/CD Job Workflows

Users Auditlog

```
mermaid
graph TB
```

```
subgraph Identity GitLab Repositories
subgraph accessctl-manifests Repo
direction TB
REPOINVMANIFESTSUSERS[("manifests/users/<br />users.yml/json/csv")]:::sky
REPOINVMANIFESTSATTRIBUTES[("manifests/attributes/<br />{attribute}.yml/json/csv")]
REPOINVMANIFESTSROLES[("manifests/roles/<br />{role}.yml/json/csv")]
REPOINVMANIFESTSOU[("manifests/ou/<br />{ou}.yml/json/csv")]
end
end
```

```
subgraph Identity Platform CI/CD Auditlog Stage Scripts
direction LR
```

```
CIUSERJOB[["Stage 2.1<br />Users Job<br/>CLI auditlog:users"]]:::orange
CIUSERPARSEMANIFESTCURRENT[(Parse Current Manifest Commit)]
CIUSERPARSEMANIFESTPREVIOUS[(Parse Previous Manifest Commit)]
CIUSERPARSEMANIFESTFILTER{Filter results<br />with differences}
CIUSERONBOARDINGCHECK{{Check if user has been created}}
CIUSEROFFBOARDINGCHECK{{Check if user has been deprovisioned}}
CIUSERATTRIBUTECHECK{{Loop through users and<br />compare attributes for differences}}

CIUSERLOGS3{{Create Audit Log entry in S3 bucket}}
CIUSERAPI{{Audit Transaction<br />REST API Call to accessctl<br />for automation workflows}}
CIUSERLOGREPO{{Append Auditlog in Repository}}

CIUSERJOB --> CIUSERPARSEMANIFESTCURRENT
CIUSERJOB --> CIUSERPARSEMANIFESTPREVIOUS
CIUSERPARSEMANIFESTCURRENT --> CIUSERPARSEMANIFESTFILTER
```

CIUSERPARSEMANIFESTPREVIOUS --> CIUSERPARSEMANIFESTFILTER

CIUSERPARSEMANIFESTFILTER --> CIUSERONBOARDINGCHECK
CIUSERPARSEMANIFESTFILTER --> CIUSEROFFBOARDINGCHECK
CIUSERPARSEMANIFESTFILTER --> CIUSERATTRIBUTECHECK
CIUSERONBOARDINGCHECK --> CIUSERLOGS3
CIUSEROFFBOARDINGCHECK --> CIUSERLOGS3
CIUSERATTRIBUTECHECK --> CIUSERLOGS3
CIUSERLOGS3 --> CIUSERAPI
CIUSERAPI --> CIUSERLOGREPO

end

subgraph Identity GitLab Repositories

subgraph accessctl-auditlog Repo

direction TB

REPOINVCHANGELOGUSERSONBOARDING[("auditlog/users/
onboarding.yml/json/csv")]:::emerald

REPOINVCHANGELOGUSERSOFFBOARDING[("auditlog/users/
offboarding.yml/json/csv")]:::emerald

REPOINVCHANGELOGUSERSCHANGES[("auditlog/users/
attributes.yml/json/csv")]:::emerald

end

end

REPOINVMANIFESTSUSERS --> CIUSERJOB

CIUSERLOGREPO --> REPOINVCHANGELOGUSERSONBOARDING

CIUSERLOGREPO --> REPOINVCHANGELOGUSERSOFFBOARDING

CIUSERLOGREPO --> REPOINVCHANGELOGUSERSCHANGES

classDef slate fill:cbd5e1,stroke:475569,stroke-width:1px;
classDef red fill:fca5a5,stroke:dc2626,stroke-width:1px;
classDef orange fill:fd8b74,stroke:ea580c,stroke-width:1px;
classDef yellow fill:fcd34d,stroke:ca8a04,stroke-width:1px;
classDef emerald fill:6ee7b7,stroke:059669,stroke-width:1px;
classDef cyan fill:67e8f9,stroke:0891b2,stroke-width:1px;
classDef sky fill:7dd3fc,stroke:0284c7,stroke-width:1px;
classDef violet fill:c4b5fd,stroke:7c3aed,stroke-width:1px;
classDef fuchsia fill:f0abfc,stroke:c026d3,stroke-width:1px;

Attributes Auditlog

mermaid

graph TB

subgraph Identity GitLab Repositories

subgraph accessctl-manifests Repo

direction TB

REPOINVMANIFESTSUSERS[("manifests/users/
users.yml/json/csv")]

REPOINVMANIFESTSATTRIBUTES[("manifests/attributes/
{attribute}.yml/json/csv")]:::sky

REPOINVMANIFESTSROLES[("manifests/roles/
{role}.yml/json/csv")]

```
REPOINVMANIFESTSOU[("manifests/ou/<br />{ou}.yaml/json/csv")]
end
end
```

subgraph Identity Platform CI/CD Auditlog Stage Scripts

direction LR

```
CIATTRIBUTEJOB[["Stage 2.2<br />Attributes Job<br/>CLI auditlog:attributes"]]:::orange
CIATTRIBUTEPARSEMANIFESTCURRENT[(Parse Current Manifest Commit)]
CIATTRIBUTEPARSEMANIFESTPREVIOUS[(Parse Previous Manifest Commit)]
CIATTRIBUTEPARSEMANIFESTFILTER{Filter results<br />with differences}
CIATTRIBUTECREATEDCHECK{{Check if attribute has been created}}
CIATTRIBUTEDELETEDCHECK{{Check if attribute has been deleted}}
CIATTRIBUTELOGS3{{Create Audit Log entry in S3 bucket}}
CIATTRIBUTEAPI{{Audit Transaction<br />REST API Call to accessctl<br />for automation workflows}}
CIATTRIBUTELOGREPO{{Append Auditlog in Repository}}
CIATTRIBUTEJOB --> CIATTRIBUTEPARSEMANIFESTCURRENT
CIATTRIBUTEJOB --> CIATTRIBUTEPARSEMANIFESTPREVIOUS
CIATTRIBUTEPARSEMANIFESTCURRENT --> CIATTRIBUTEPARSEMANIFESTFILTER
CIATTRIBUTEPARSEMANIFESTPREVIOUS --> CIATTRIBUTEPARSEMANIFESTFILTER
CIATTRIBUTEPARSEMANIFESTFILTER --> CIATTRIBUTECREATEDCHECK
CIATTRIBUTEPARSEMANIFESTFILTER --> CIATTRIBUTEDELETEDCHECK
CIATTRIBUTECREATEDCHECK --> CIATTRIBUTELOGS3
CIATTRIBUTEDELETEDCHECK --> CIATTRIBUTELOGS3
CIATTRIBUTELOGS3 --> CIATTRIBUTEAPI
CIATTRIBUTEAPI --> CIATTRIBUTELOGREPO
end
```

subgraph Identity GitLab Repositories

subgraph accessctl-auditlog Repo

direction TB

```
REPOINVCHANGELOGATTRIBUTE[("auditlog/attribute/<br />{attribute}.yaml/json/csv")]:::emerald
REPOINVCHANGELOGROLE[("auditlog/role/<br />{role}.yaml/json/csv")]
REPOINVCHANGELOGOU[("auditlog/ou/<br />{ou}.yaml/json/csv")]
end
end
```

REPOINVMANIFESTSATTRIBUTES --> CIATTRIBUTEJOB

CIATTRIBUTELOGREPO --> REPOINVCHANGELOGATTRIBUTE

```
classDef slate fill:cbd5e1,stroke:475569,stroke-width:1px;
classDef red fill:fca5a5,stroke:dc2626,stroke-width:1px;
classDef orange fill:fdba74,stroke:ea580c,stroke-width:1px;
classDef yellow fill:fcd34d,stroke:ca8a04,stroke-width:1px;
classDef emerald fill:6ee7b7,stroke:059669,stroke-width:1px;
classDef cyan fill:67e8f9,stroke:0891b2,stroke-width:1px;
classDef sky fill:7dd3fc,stroke:0284c7,stroke-width:1px;
classDef violet fill:c4b5fd,stroke:7c3aed,stroke-width:1px;
classDef fuchsia fill:f0abfc,stroke:c026d3,stroke-width:1px;
```

Roles Auditlog

mermaid
graph TB

subgraph Identity GitLab Repositories

subgraph accessctl-manifests Repo

direction TB

REPOINVMANIFESTSUSERS[("manifests/users/
users.yml/json/csv")]

REPOINVMANIFESTSATTRIBUTES[("manifests/attributes/
{attribute}.yml/json/csv")]

REPOINVMANIFESTSROLES[("manifests/roles/
{role}.yml/json/csv")]:::sky

REPOINVMANIFESTSOU[("manifests/ou/
{ou}.yml/json/csv")]

end

end

subgraph Identity Platform CI/CD Auditlog Stage Scripts

direction LR

CIROLEJOB[["Stage 2.3
Roles Job
CLI auditlog:roles"]]:::orange

CIROLEPARSEMANIFESTCURRENT[(Parse Current Manifest Commit)]

CIROLEPARSEMANIFESTPREVIOUS[(Parse Previous Manifest Commit)]

CIROLEPARSEMANIFESTFILTER{Filter results
with differences}

CIROLECREATEDCHECK{{Check if user has been added}}

CIROLEDELETEDCHECK{{Check if user has been deleted}}

CIROLELOGS3{{Create Audit Log entry in S3 bucket}}

CIROLEAPI{{Audit Transaction
REST API Call to accessctl
for automation workflows}}

CIROLELOGREPO{{Append Auditlog in Repository}}

CIROLEJOB --> CIROLEPARSEMANIFESTCURRENT

CIROLEJOB --> CIROLEPARSEMANIFESTPREVIOUS

CIROLEPARSEMANIFESTCURRENT --> CIROLEPARSEMANIFESTFILTER

CIROLEPARSEMANIFESTPREVIOUS --> CIROLEPARSEMANIFESTFILTER

CIROLEPARSEMANIFESTFILTER --> CIROLECREATEDCHECK

CIROLEPARSEMANIFESTFILTER --> CIROLEDELETEDCHECK

CIROLECREATEDCHECK --> CIROLELOGS3

CIROLEDELETEDCHECK --> CIROLELOGS3

CIROLELOGS3 --> CIROLEAPI

CIROLEAPI --> CIROLELOGREPO

end

subgraph Identity GitLab Repositories

subgraph accessctl-auditlog Repo

direction TB

REPOINVCHANGELOGATTRIBUTE[("auditlog/attribute/
{attribute}.yml/json/csv")]

REPOINVCHANGELOGROLE[("auditlog/role/
{role}.yml/json/csv")]:::emerald

REPOINVCHANGELOGOU[("auditlog/ou/
{ou}.yml/json/csv")]

end

end

REPOINVMANIFESTSROLES --> CIROLEJOB

CIROLELOGREPO --> REPOINVCHANGELOGROLE

```

classDef slate fill:cbd5e1,stroke:475569,stroke-width:1px;
classDef red fill:fca5a5,stroke:dc2626,stroke-width:1px;
classDef orange fill:fd8b74,stroke:ea580c,stroke-width:1px;
classDef yellow fill:fcd34d,stroke:ca8a04,stroke-width:1px;
classDef emerald fill:6ee7b7,stroke:059669,stroke-width:1px;
classDef cyan fill:67e8f9,stroke:0891b2,stroke-width:1px;
classDef sky fill:7dd3fc,stroke:0284c7,stroke-width:1px;
classDef violet fill:c4b5fd,stroke:7c3aed,stroke-width:1px;
classDef fuchsia fill:f0abfc,stroke:c026d3,stroke-width:1px;

```

Organization Units Auditlog

```

mermaid
graph TB

```

```

subgraph Identity GitLab Repositories

```

```

subgraph accessctl-manifests Repo

```

```

direction TB

```

```

REPOINVMANIFESTSUSERS[("manifests/users/<br />users.yml/json/csv")]

```

```

REPOINVMANIFESTSATTRIBUTES[("manifests/attributes/<br />{attribute}.yml/json/csv")]

```

```

REPOINVMANIFESTSROLES[("manifests/roles/<br />{role}.yml/json/csv")]

```

```

REPOINVMANIFESTSOU[("manifests/ou/<br />{ou}.yml/json/csv")]:::sky

```

```

end

```

```

end

```

```

subgraph Identity Platform CI/CD Auditlog Stage Scripts

```

```

direction LR

```

```

CIGROUPJOB[["Stage 2.4<br />Org Units Job<br/>CLI auditlog:ou"]]:::orange

```

```

CIGROUPPARSEMANIFESTCURRENT[(Parse Current Manifest Commit)]

```

```

CIGROUPPARSEMANIFESTPREVIOUS[(Parse Previous Manifest Commit)]

```

```

CIGROUPPARSEMANIFESTFILTER{Filter results<br />with differences}

```

```

CIGROUPCREATEDCHECK{{Check if user has been added}}

```

```

CIGROUPDELETEDCHECK{{Check if user has been deleted}}

```

```

CIGROUPLOGS3{{Create Audit Log entry in S3 bucket}}

```

```

CIGROUPAPI{{Audit Transaction<br />REST API Call to accessctl<br />for automation workflows}}

```

```

CIGROUPLOGREPO{{Append Auditlog in Repository}}

```

```

CIGROUPJOB --> CIGROUPPARSEMANIFESTCURRENT

```

```

CIGROUPJOB --> CIGROUPPARSEMANIFESTPREVIOUS

```

```

CIGROUPPARSEMANIFESTCURRENT --> CIGROUPPARSEMANIFESTFILTER

```

```

CIGROUPPARSEMANIFESTPREVIOUS --> CIGROUPPARSEMANIFESTFILTER

```

```

CIGROUPPARSEMANIFESTFILTER --> CIGROUPCREATEDCHECK

```

```

CIGROUPPARSEMANIFESTFILTER --> CIGROUPDELETEDCHECK

```

```

CIGROUPCREATEDCHECK --> CIGROUPLOGS3

```

```

CIGROUPDELETEDCHECK --> CIGROUPLOGS3

```

```

CIGROUPLOGS3 --> CIGROUPAPI

```

```

CIGROUPAPI --> CIGROUPLOGREPO

```

```

end

```

```

subgraph Identity GitLab Repositories

```

```

subgraph accessctl-auditlog Repo
direction TB
REPOINVCHANGELOGATTRIBUTE[("auditlog/attribute/<br />{attribute}.yaml/json/csv")]
REPOINVCHANGELOGROLE[("auditlog/role/<br />{role}.yaml/json/csv")]
REPOINVCHANGELOGOU[("auditlog/ou/<br />{ou}.yaml/json/csv")]:::emerald
end
end

```

```

REPOINVMANIFESTSOU --> CIGROUPJOB
CIGROUPLOGREPO --> REPOINVCHANGELOGOU

```

```

classDef slate fill:cbd5e1,stroke:475569,stroke-width:1px;
classDef red fill:fca5a5,stroke:dc2626,stroke-width:1px;
classDef orange fill:fdba74,stroke:ea580c,stroke-width:1px;
classDef yellow fill:fcd34d,stroke:ca8a04,stroke-width:1px;
classDef emerald fill:6ee7b7,stroke:059669,stroke-width:1px;
classDef cyan fill:67e8f9,stroke:0891b2,stroke-width:1px;
classDef sky fill:7dd3fc,stroke:0284c7,stroke-width:1px;
classDef violet fill:c4b5fd,stroke:7c3aed,stroke-width:1px;
classDef fuchsia fill:f0abfc,stroke:c026d3,stroke-width:1px;

```

SECTION: security/identity/platform/manifests/_index.md

{{% alert title="Not Live Yet" color="warning" %}}

You are viewing a preview of documentation for the future state of GitLab Identity v3 (mid 2024). See the [Access Management Policy](/handbook/security/security-and-technology-policies/access-management-policy/) for the GitLab Identity v2 current state with baseline entitlements and access requests. See the roadmap in the [epics gantt chart](https://gitlab.com/groups/gitlab-com/gf-security/identity/eng/-/roadmap?state=all&sort=startdateasc&layout=QUARTERS&timeframerangetype=THREEYEARS&groupparent=gitlab-com/gf-security/identity/eng&progress=WEIGHT&showprogress=true&showmilestones=false&milestonetype=ALL&showlabels=true).

{{% /alert %}}

Pipeline Overview

```

mermaid
graph LR

```

```

subgraph accessctl GitLab Repositories
direction LR
subgraph accessctl-policies Repo
direction TB
REPOINVPOLICIESROLES[("policies/role/{kingdom}.yaml")]:::sky
REPOINVPOLICIESOU[("policies/ou/{kingdom}.yaml")]:::sky
end
subgraph accessctl-manifests Repo
direction TB

```

```

REPOINVMANIFESTSUSERS[("manifests/users/<br />users.yml/json/csv")]:::emerald
REPOINVMANIFESTSATTRIBUTES[("manifests/attributes/<br />{attribute}.yml/json/csv")]:::emerald
REPOINVMANIFESTSROLE[("manifests/roles/<br />{role}.yml/json/csv")]:::emerald
REPOINVMANIFESTSOU[("manifests/ou/<br />{ou}.yml/json/csv")]:::emerald
end
subgraph accessctl-auditlog Repo
direction TB
REPOINVCHANGELOGUSERSONBOARDING[("auditlog/users/<br />onboarding.yml/json/csv")]
REPOINVCHANGELOGUSERSOFFBOARDING[("auditlog/users/<br />offboarding.yml/json/csv")]
REPOINVCHANGELOGUSERSCHANGES[("auditlog/users/<br />attributes.yml/json/csv")]
REPOINVCHANGELOGATTRIBUTE[("auditlog/attribute/<br />{attribute}.yml/json/csv")]
REPOINVCHANGELOGROLE[("auditlog/role/<br />{role}.yml/json/csv")]
REPOINVCHANGELOGOU[("auditlog/ou/<br />{ou}.yml/json/csv")]
end
end

```

```

subgraph accessctl GitLab CI/CD Pipeline Jobs

```

```

direction LR

```

```

subgraph Manifest Stage

```

```

direction LR

```

```

CIMANIFESTUSERJOB[["Stage 1.1<br />Users Job<br/>CLI manifest:users"]]:::orange

```

```

CIMANIFESTROLEJOB[["Stage 1.2<br />Roles Job<br />CLI manifest:roles"]]:::orange

```

```

CIMANIFESTGROUPJOB[["Stage 1.3<br />Org Units Job<br/>CLI manifest:ou"]]:::orange

```

```

end

```

```

subgraph Auditlog Stage

```

```

direction LR

```

```

CICHANGELOGUSERJOB[["Stage 2.1<br />Users Job<br/>CLI auditlog:users"]]

```

```

CICHANGELOGATTRIBUTEJOB[["Stage 2.2<br />Attributes Job<br/>CLI auditlog:attributes"]]

```

```

CICHANGELOGROLEJOB[["Stage 2.3<br />Roles Job<br/>CLI auditlog:roles"]]

```

```

CICHANGELOGGROUPJOB[["Stage 2.4<br />Org Units Job<br/>CLI auditlog:ou"]]

```

```

end

```

```

subgraph Provisioning Stage

```

```

direction LR

```

```

subgraph Okta API

```

```

CIPROVISIONINGOKTAUSERJOB["Stage 3.1<br />Okta Users Job<br />provision:okta-users"]

```

```

CIPROVISIONINGOKTAGROUPJOB["Stage 3.2<br />Okta Groups Job<br />provision:okta-groups"]

```

```

end

```

```

subgraph Google Workspace Directory API

```

```

CIPROVISIONINGGOOGLEGROUPJOB["Stage 3.3<br />Google Groups Job<br />provision:google-groups"]

```

```

end

```

```

subgraph GitLab.com SaaS API

```

```

CIPROVISIONINGGITLABSAASGROUPJOB["Stage 3.4<br />GitLab SaaS Groups Job<br />provision:gitlab-saas-groups"]

```

```

end

```

```

subgraph GitLab Self-Managed Instance API

```

```

CIPROVISIONINGGITLABSELFGROUPJOB["Stage 3.5<br />GitLab Self-Managed Groups Job<br />provision:gitlab-self-groups"]

```

```

end

```

```

end

```

CIMANIFESTUSERJOB --> CIMANIFESTROLEJOB --> CIMANIFESTGROUPJOB
CICHANGELOGUSERJOB --> CICHANGELOGATTRIBUTEJOB --> CICHANGELOGROLEJOB --> CICHANGELOGGROUPJOB
CIPROVISIONINGOKTAUSERJOB --> CIPROVISIONINGOKTAGROUPJOB --> CIPROVISIONINGGOOGLEGROUPJOB
CIPROVISIONINGGITLABSAASGROUPJOB --> CIPROVISIONINGGITLABSELFGROUPJOB

click CIMANIFESTUSERJOB "/handbook/security/identity/platform/manifests" "View Details"
click CIMANIFESTROLEJOB "/handbook/security/identity/platform/manifests" "View Details"
click CIMANIFESTGROUPJOB "/handbook/security/identity/platform/manifests" "View Details"
click CICHANGELOGUSERJOB "/handbook/security/identity/platform/auditlog" "View Details"
click CICHANGELOGATTRIBUTEJOB "/handbook/security/identity/platform/auditlog" "View Details"
click CICHANGELOGROLEJOB "/handbook/security/identity/platform/auditlog" "View Details"
click CICHANGELOGGROUPJOB "/handbook/security/identity/platform/auditlog" "View Details"
click CIPROVISIONINGOKTAUSERJOB "/handbook/security/identity/platform/provisioning/okta" "View Details"
click CIPROVISIONINGOKTAGROUPJOB "/handbook/security/identity/platform/provisioning/okta" "View Details"
click CIPROVISIONINGGOOGLEGROUPJOB "/handbook/security/identity/platform/provisioning/google" "View Details"
click CIPROVISIONINGGITLABSAASGROUPJOB
"/handbook/security/identity/platform/provisioning/gitlab" "View Details"
click CIPROVISIONINGGITLABSELFGROUPJOB
"/handbook/security/identity/platform/provisioning/gitlab" "View Details"

```
classDef slate fill:cbd5e1,stroke:475569,stroke-width:1px;  
classDef red fill:fca5a5,stroke:dc2626,stroke-width:1px;  
classDef orange fill:fdba74,stroke:ea580c,stroke-width:1px;  
classDef yellow fill:fcd34d,stroke:ca8a04,stroke-width:1px;  
classDef emerald fill:6ee7b7,stroke:059669,stroke-width:1px;  
classDef cyan fill:67e8f9,stroke:0891b2,stroke-width:1px;  
classDef sky fill:7dd3fc,stroke:0284c7,stroke-width:1px;  
classDef violet fill:c4b5fd,stroke:7c3aed,stroke-width:1px;  
classDef fuchsia fill:f0abfc,stroke:c026d3,stroke-width:1px;  
end
```

CI/CD Job Workflows

mermaid
graph TB

```
subgraph "Workday"  
    WORKDAYWORKER[Worker]  
    WORKDAYREGION([Region])  
    WORKDAYDEPARTMENT([Department])  
    WORKDAYDIVISION([Division])  
    WORKDAYJOBTITLE([Job Title])  
    WORKDAYDEPARTMENT -. WORKDAYWORKER  
    WORKDAYDIVISION -. WORKDAYWORKER  
    WORKDAYREGION -. WORKDAYWORKER  
    WORKDAYJOBTITLE -. WORKDAYWORKER  
end
```

```
    WORKDAYWORKER -. "Manager Relationship" .-> WORKDAYWORKER
end
```

```
subgraph Identity SaaS Vendor Services
```

```
direction LR
```

```
subgraph Okta GitLab Tenant
```

```
OKTAPROFILEMAPPING{{Workday to Okta<br>Profile Mapping}}
```

```
IDENTITYVENDOROKTAAPIUSERS(["Okta Users"])
```

```
IDENTITYVENDOROKTAAPIENDPOINT(["Okta API<br />https://gitlab.okta.com/api/v1"])
```

```
WORKDAYWORKER -- "SCIM Provisioning<br />Every Hour" --> OKTAPROFILEMAPPING
```

```
OKTAPROFILEMAPPING --> IDENTITYVENDOROKTAAPIUSERS
```

```
IDENTITYVENDOROKTAAPIUSERS --> IDENTITYVENDOROKTAAPIENDPOINT
```

```
end
```

```
end
```

```
subgraph Identity Platform CI/CD Manifest Stage Scripts
```

```
direction LR
```

```
CIUSERJOB[["Stage 1.1<br />Users Job<br>CLI manifest:users"]]:::orange
```

```
CIROLEJOB[["Stage 1.2<br />Roles Job<br>CLI manifest:roles"]]:::orange
```

```
CIGROUPJOB[["Stage 1.3<br />Org Units Job<br>CLI manifest:ou"]]:::orange
```

```
CIUSERPARSEAPI[(Get All Users<br>from Okta API)]
```

```
CIUSERTRANSFORM{{Transform users with<br>snakecase attributes}}
```

```
CIUSERMANIFESTUSERS{{Create manifest of users}}
```

```
CIUSERMANIFESTATTRIBUTES{{Create manifest of unique<br />attributes and count of users}}
```

```
CIUSERJOB --> CIUSERPARSEAPI
```

```
CIUSERPARSEAPI --> CIUSERTRANSFORM
```

```
CIUSERTRANSFORM --> CIUSERMANIFESTUSERS
```

```
CIUSERMANIFESTUSERS ----> CIUSERMANIFESTATTRIBUTES
```

```
CIROLEPARSEPOLICY[(Parse Users<br />Parse Attributes<br />Parse Policies<br />Parse Policy Rules)]
```

```
CIROLEVALIDATEKEYS{{Validate attribute keys exist<br />for each policy rule}}
```

```
CIROLEGETNAMEDUSERS{{Get users<br />for each policy handle}}
```

```
CIROLEGETUSERSFORATTRIBUTE{{Get users<br />for each Attribute Key}}
```

```
CIROLECREATEMANIFEST{{Create manifest of users<br />with unique email addresses<br />for each role}}
```

```
CIROLEJOB --> CIROLEPARSEPOLICY
```

```
CIROLEPARSEPOLICY --> CIROLEVALIDATEKEYS
```

```
CIROLEVALIDATEKEYS --> CIROLEGETNAMEDUSERS
```

```
CIROLEGETNAMEDUSERS --> CIROLEGETUSERSFORATTRIBUTE
```

```
CIROLEGETUSERSFORATTRIBUTE ---> CIROLECREATEMANIFEST
```

```
CIGROUPPARSEPOLICY[(Parse Users<br />Parse Roles<br />Parse Attributes<br />Parse Policies<br />Parse Policy Rules)]
```

```
CIGROUPVALIDATEKEYS{{Validate attribute keys<br />and role keys exist<br />for each policy rule}}
```

```
CIGROUPGETNAMEDUSERS{{Get users<br />for each policy handle}}
```

```
CIGROUPGETUSERSFORROLE{{Get users<br />for each Role Key}}
```

```
CIGROUPGETUSERSFORDIMENSION{{Get users<br />for each Attribute Key}}
```

```

CIGROUPCREATEMANIFEST{{Create manifest<br />with unique email addresses<br />for each group}}
CIGROUPJOB --> CIGROUPPARSEPOLICY
CIGROUPPARSEPOLICY --> CIGROUPVALIDATEKEYS
CIGROUPVALIDATEKEYS --> CIGROUPGETNAMEDUSERS
CIGROUPGETNAMEDUSERS --> CIGROUPGETUSERSFORROLE
CIGROUPGETUSERSFORROLE --> CIGROUPGETUSERSFORDIMENSION
CIGROUPGETUSERSFORDIMENSION --> CIGROUPCREATEMANIFEST
end

```

```

subgraph Identity GitLab Repositories
subgraph accessctl-policies Repo
direction LR
REPOINVPOLICIESROLES[("policies/role/{kingdom}.yaml"):::sky
REPOINVPOLICIESOU[("policies/ou/{kingdom}.yaml"):::sky
end
end

```

```

subgraph Identity GitLab Repositories
subgraph accessctl-inventory Repo
direction TB
REPOINVMANIFESTSUSERS[("manifests/users/<br />users.yaml/json/csv"):::emerald
REPOINVMANIFESTSATTRIBUTES[("manifests/attributes/<br />{attribute}.yaml/json/csv"):::emerald
REPOINVMANIFESTSROLE[("manifests/roles/<br />{role}.yaml/json/csv"):::emerald
REPOINVMANIFESTSOU[("manifests/ou/<br />{ou}.yaml/json/csv"):::emerald
end
end

```

```

CIUSERMANIFESTUSERS --> REPOINVMANIFESTSUSERS
CIUSERMANIFESTATTRIBUTES --> REPOINVMANIFESTSATTRIBUTES

```

```

IDENTITYVENDOROKTAAPIENDPOINT --> CIUSERJOB
REPOINVPOLICIESROLES --> CIROLEJOB
REPOINVPOLICIESOU --> CIGROUPJOB
CIROLECREATEMANIFEST --> REPOINVMANIFESTSROLE
CIGROUPCREATEMANIFEST --> REPOINVMANIFESTSGROUPS

```

```

classDef slate fill:cbd5e1,stroke:475569,stroke-width:1px;
classDef red fill:fca5a5,stroke:dc2626,stroke-width:1px;
classDef orange fill:fdba74,stroke:ea580c,stroke-width:1px;
classDef yellow fill:fcd34d,stroke:ca8a04,stroke-width:1px;
classDef emerald fill:6ee7b7,stroke:059669,stroke-width:1px;
classDef cyan fill:67e8f9,stroke:0891b2,stroke-width:1px;
classDef sky fill:7dd3fc,stroke:0284c7,stroke-width:1px;
classDef violet fill:c4b5fd,stroke:7c3aed,stroke-width:1px;
classDef fuchsia fill:f0abfc,stroke:c026d3,stroke-width:1px;

```

Policy and Manifest Breaking Changes

Attribute Key No Longer Exists

When a policy is parsed prior to creating a manifest, the manifests/attributes/{attribute}.yaml file is parsed to verify that the attribute key exists. This file includes the latest list of unique values from users in the Okta API.

This will detect if a department, division, title, etc no longer exists (ex. has been renamed upstream).

If a policy uses an attribute key value that no longer exists, no updated manifest will be created so the current (previous) state will be retained. This freezes the current manifest until the policy is updated by Identity Operations and the policy CODEOWNERS.

See the updating keys in policies documentation to learn how this process has been partially automated.

Manager No Longer Exists

Similar to an attribute key that no longer exists, every manager value is verified against the latest manifest of users. This will ensure that the user has not been offboarded and email handle hasn't changed (ex. maiden name change).

If the manager user no longer exists based on the manager handle defined in the policy, no updated manifest will be created so the current (previous) state will be retained. This freezes the current manifest until the policy is updated by Identity Operations and the policy CODEOWNERS.

See the updating keys in policies documentation to learn how this process has been partially automated.

Updating Keys in Policies

If a key in a policy cannot be found, a branch and merge request is automatically created by accessctl and the CODEOWNERS and Identity Operations team members are assigned to make edits.

A comment is posted automatically to the merge request with the list of the previous manifest users and their latest updated values for the key of the value that does not exist anymore. This automates the research work to determine what the new values appear to be upstream that can be verified or adjusted by CODEOWNERS.

The policy manifest remains frozen until the changes are made. After the branch is merged, an updated manifest will be created on the next pipeline run which automatically heals the problem.

SECTION: security/identity/platform/provisioning/_index.md

{{% alert title="Work in Progress" color="danger" %}}

This page is a work-in-progress. Please check back later for up-to-date details.

{{% /alert %}}

See the specific pages for each of our API clients.

SECTION: security/identity/platform/provisioning/gitlab/_index.md

{{% alert title="Not Live Yet" color="warning" %}}

You are viewing a preview of documentation for the future state of GitLab Identity v3 (mid 2024). See the [Access Management Policy](/handbook/security/security-and-technology-policies/access-management-policy/) for the GitLab Identity v2 current state with baseline entitlements and access requests. See the roadmap in the [epics gantt chart](https://gitlab.com/groups/gitlab-com/gl-security/identity/eng/-/roadmap?state=all&sort=startdateasc&layout=QUARTERS&timeframerangetype=THREEYEARS&groupparent=gitlab-com/gl-security/identity/eng&progress=WEIGHT&showprogress=true&showmilestones=false&milestonetype=ALL&showlabels=true).

{{% /alert %}}

{{% alert title="Work in Progress" color="danger" %}}

This page is a work-in-progress. Please check back later for up-to-date details.

{{% /alert %}}

SECTION: security/identity/platform/provisioning/google/_index.md

{{% alert title="Not Live Yet" color="warning" %}}

You are viewing a preview of documentation for the future state of GitLab Identity v3 (mid 2024). See the [Access Management Policy](/handbook/security/security-and-technology-policies/access-management-policy/) for the GitLab Identity v2 current state with baseline entitlements and access requests. See the roadmap in the [epics gantt chart](https://gitlab.com/groups/gitlab-com/gl-security/identity/eng/-/roadmap?state=all&sort=startdateasc&layout=QUARTERS&timeframerangetype=THREEYEARS&groupparent=gitlab-com/gl-security/identity/eng&progress=WEIGHT&showprogress=true&showmilestones=false&milestonetype=ALL&showlabels=true).

{{% /alert %}}

{{% alert title="Work in Progress" color="danger" %}}

This page is a work-in-progress. Please check back later for up-to-date details.

{{% /alert %}}

SECTION: security/identity/platform/provisioning/okta/_index.md

{{% alert title="Not Live Yet" color="warning" %}}

You are viewing a preview of documentation for the future state of GitLab Identity v3 (mid 2024). See the [Access Management Policy](/handbook/security/security-and-technology-policies/access-management-policy/) for the GitLab Identity v2 current state with baseline entitlements and access requests. See the roadmap in the [epics gantt chart](https://gitlab.com/groups/gitlab-com/gl-security/identity/eng/-/roadmap?state=all&sort=startdateasc&layout=QUARTERS&timeframerangetype=THREEYEARS&groupparent=gitlab-com/gl-security/identity/eng&progress=WEIGHT&showprogress=true&showmilestones=false&milestonetype=ALL&showlabels=true).

/roadmap?state=all&sort=startdateasc&layout=QUARTERS&timeframerangetype=THREEYEARS&groupptlab-com/gl-security/identity/eng&progress=WEIGHT&showprogress=true&showmilestones=false&milestonestype=ALL&showlabels=true">epics gantt chart.

{{% /alert %}}

{{% alert title="GitOps Architecture and Approval Rules" color="info" %}}

This page is specific to the Okta groups and users configuration. You may also be interested in the Okta GitOps Terraform architecture for applications and settings configuration and merge request approvals documentation.

{{% /alert %}}

Pipeline Overview

mermaid
graph LR

```
subgraph accessctl GitLab Repositories
  direction LR
  subgraph accessctl-policies Repo
    direction TB
    REPOINVPOLICIESROLES[("policies/role/{kingdom}.yaml")]
    REPOINVPOLICIESOU[("policies/ou/{kingdom}.yaml")]
  end
  subgraph accessctl-inventory Repo
    direction LR
    subgraph manifests
      direction TB
      REPOINVMANIFESTSUSERS[("manifests/users/<br />users.yaml/json/csv")]:::sky
      REPOINVMANIFESTSATTRIBUTES[("manifests/attributes/<br />{attribute}.yaml/json/csv")]
      REPOINVMANIFESTSROLE[("manifests/roles/<br />{role}.yaml/json/csv")]:::sky
      REPOINVMANIFESTSOU[("manifests/ou/<br />{ou}.yaml/json/csv")]:::sky
    end
  end
end
subgraph au
```